# HEART DISEASE PREDICTION USING MACHINE LEARNING

## Mini Project Documentation

### Submitted By

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**Project Type:** Mini Machine Learning Project
**Domain:** Healthcare Analytics
**Technology Used:** Python, Machine Learning, Data Analysis

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# CERTIFICATE

This is to certify that the project titled **"Heart Disease Prediction Using Machine Learning"** is a genuine work completed as part of the mini project requirement. The project focuses on predicting heart disease using machine learning algorithms and data analysis techniques.

The project demonstrates the practical implementation of machine learning concepts such as data preprocessing, exploratory data analysis, feature scaling, model building, model evaluation, and prediction.

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# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this mini project. I thank my guide, faculty members, friends, and family members for their continuous support and encouragement.

I also thank the open-source community and Kaggle platform for providing the dataset required for this project.

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# ABSTRACT

Heart disease is one of the leading causes of death worldwide. Early prediction of heart disease can help doctors and patients take preventive actions and reduce health risks. This project aims to develop a machine learning model capable of predicting the presence of heart disease using patient medical information.

The project uses a heart disease dataset collected from Kaggle. The dataset contains several medical attributes such as age, chest pain type, cholesterol level, blood pressure, heart rate, fasting blood sugar, and other clinical measurements.

Different stages of machine learning are implemented in this project including:

- Data collection
- Data preprocessing
- Handling missing values
- Outlier detection and treatment
- Exploratory Data Analysis (EDA)
- Feature scaling
- Model training
- Model evaluation
- Prediction analysis

The project successfully demonstrates how machine learning techniques can assist in healthcare prediction systems.

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# 1. INTRODUCTION

Machine Learning is one of the most important technologies in the modern world. It helps computers learn patterns from data and make predictions without being explicitly programmed.

Healthcare is one of the major fields where machine learning is widely used. Predicting diseases using machine learning helps doctors make faster and more accurate decisions.

Heart disease is a serious medical condition affecting millions of people globally. Many factors such as age, blood pressure, cholesterol level, heart rate, and chest pain contribute to heart disease.

The main purpose of this project is to build a machine learning system that predicts whether a person has heart disease or not using medical parameters.

The project uses Python programming language along with machine learning libraries such as:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

The project workflow includes data preprocessing, visualization, feature scaling, model training, and evaluation.

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# 2. PROBLEM STATEMENT

Heart disease prediction is an important healthcare problem. Many patients are diagnosed only after symptoms become severe. Manual diagnosis may take more time and sometimes may not provide accurate results.

The problem addressed in this project is:

**“To develop a machine learning model that predicts heart disease based on patient medical information.”**

The project focuses on analyzing medical data and identifying patterns related to heart disease.

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# 3. OBJECTIVES

The major objectives of this project are:

- To study the heart disease dataset.
- To preprocess the dataset for machine learning.
- To identify important medical features.
- To perform exploratory data analysis.
- To detect and handle outliers.
- To scale numerical features.
- To build a machine learning prediction model.
- To evaluate model performance.
- To predict whether a patient has heart disease.

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# 4. SCOPE OF THE PROJECT

The project has wide applications in healthcare analytics. The scope includes:

- Early heart disease prediction.
- Supporting doctors in decision making.
- Reducing healthcare risks.
- Improving medical diagnosis systems.
- Assisting healthcare researchers.

This project can be further integrated into hospital management systems and healthcare applications.

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# 5. LITERATURE SURVEY

Several researchers have used machine learning techniques for disease prediction. Heart disease prediction is one of the most researched areas in healthcare analytics.

Traditional diagnosis methods depend on doctor experience and medical tests. Machine learning models improve prediction accuracy by analyzing large amounts of medical data.

Different algorithms commonly used for heart disease prediction include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Naive Bayes

Researchers have found that proper preprocessing and feature selection significantly improve model performance.

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# 6. SYSTEM REQUIREMENTS

## Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 20GB free space
- System Type: 64-bit

## Software Requirements

- Operating System: Windows/Linux/Mac
- Programming Language: Python
- IDE: Jupyter Notebook / VS Code
- Libraries Used:
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

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# 7. DATASET DESCRIPTION

The dataset used in this project is the **Heart Disease Dataset** downloaded from Kaggle.

The dataset contains patient medical records and heart-related clinical parameters.

## Important Features in Dataset

| Feature Name | Description |
| --- | --- |
| age | Age of patient |
| sex | Gender of patient |
| cp | Chest pain type |
| trestbps | Resting blood pressure |
| chol | Cholesterol level |

| Feature Name | Description |
|---|---|
| fbs | Fasting blood sugar |
| restecg | Resting ECG results |
| thalach | Maximum heart rate achieved |
| exang | Exercise induced angina |
| oldpeak | ST depression value |
| slope | Slope of peak exercise ST segment |
| ca | Number of major vessels |
| thal | Thalassemia |
| target | Heart disease prediction result |

## Target Column

The target column is:

- **0 = No Heart Disease**
- **1 = Heart Disease Present**

The dataset is suitable for classification problems in machine learning.

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# 8. METHODOLOGY

The project follows a systematic machine learning workflow.

## Step 1: Data Collection

The dataset is collected from Kaggle in CSV format.

## Step 2: Data Loading

The dataset is loaded into Python using Pandas.

```
import pandas as pd

df = pd.read_csv('heart-2.csv')
```

### Step 3: Data Preprocessing

The dataset is checked for:

- Missing values
- Duplicate values
- Incorrect data types
- Outliers

### Step 4: Exploratory Data Analysis

Visualization techniques are used to understand relationships between features.

### Step 5: Feature Scaling

StandardScaler is used to normalize numerical data.

### Step 6: Model Building

Machine learning algorithms are trained using training data.

### Step 7: Model Evaluation

Accuracy and prediction performance are evaluated.

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# 9. DATA PREPROCESSING

Data preprocessing is one of the most important steps in machine learning.

## Checking Missing Values

```
df.isnull().sum()
```

The dataset was checked for missing values.

## Checking Duplicate Values

Duplicate records were identified and removed if necessary.

### Data Cleaning

Data cleaning improves dataset quality and model performance.

### Data Transformation

Categorical values are converted into machine-readable format.

### Label Encoding

Label encoding is used for converting categorical variables into numeric form.

```
from sklearn.preprocessing import LabelEncoder
```

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## 10. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis helps understand data patterns and relationships.

### Correlation Analysis

Correlation analysis identifies relationships between variables.

```
df.corr()
```

### Visualization Techniques Used

- Heatmap
- Boxplot
- Countplot
- Histogram

### Benefits of EDA

- Identifies important features
- Detects outliers
- Finds hidden patterns
- Improves model understanding

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# 11. OUTLIER DETECTION AND HANDLING

Outliers are abnormal values that negatively affect model performance.

The project uses the IQR (Interquartile Range) method for detecting outliers.

## Outlier Detection Function

```
def replace_outliers(dataframe, feature):
    q1 = dataframe[feature].quantile(0.25)
    q3 = dataframe[feature].quantile(0.75)
    iqr = q3 - q1

    lower_range = q1 - 1.5 * iqr
    upper_range = q3 + 1.5 * iqr

    dataframe[feature] = dataframe[feature].clip(lower_range, upper_range)
```

## Features Checked for Outliers

- trestbps
- chol
- fbs
- thalach

## Boxplot Visualization

Seaborn boxplots are used for visualizing outliers.

```
sns.boxplot(x='chol', data=df)
```

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# 12. FEATURE ENGINEERING

Feature engineering improves machine learning model performance.

## Feature Scaling

The project uses StandardScaler for scaling numerical features.

```
from sklearn.preprocessing import StandardScaler
```

### Why Scaling is Important

- Improves algorithm performance
- Reduces bias due to large values
- Helps gradient-based algorithms
- Increases model accuracy

### Train-Test Split

The dataset is divided into:

- Training Data
- Testing Data

```
from sklearn.model_selection import train_test_split
```

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# 13. MODEL BUILDING

Machine learning models are trained using the processed dataset.

### Algorithms Used

The project imports several machine learning algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

### Logistic Regression

Logistic Regression is commonly used for binary classification problems.

```
from sklearn.linear_model import LogisticRegression
```

### Decision Tree

Decision Tree creates rule-based predictions.

```python
from sklearn.tree import DecisionTreeClassifier
```

### Random Forest

Random Forest improves prediction accuracy using multiple decision trees.

```python
from sklearn.ensemble import RandomForestClassifier
```

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## 14. MODEL EVALUATION

Model evaluation measures the performance of machine learning algorithms.

### Evaluation Metrics Used

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

### Accuracy Score

Accuracy measures correct predictions.

```python
from sklearn.metrics import accuracy_score
```

### Confusion Matrix

Confusion matrix shows:

- True Positive
- True Negative
- False Positive
- False Negative

### Importance of Evaluation

Evaluation helps compare model performance and select the best algorithm.

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# 15. RESULTS AND DISCUSSION

The machine learning model successfully predicts heart disease using patient medical data.

## Observations

- Feature scaling improved model performance.
- Outlier handling increased prediction stability.
- Correlation analysis identified important features.
- Logistic Regression provided effective classification results.

## Prediction Analysis

The model predicts whether a patient has heart disease based on clinical parameters.

The project demonstrates that machine learning can assist healthcare professionals in disease prediction.

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# 16. ADVANTAGES

## Advantages of the Project

- Early detection of heart disease
- Faster medical analysis
- Improved prediction accuracy
- Supports doctors in diagnosis
- Reduces manual effort
- Cost-effective healthcare solution

## Advantages of Machine Learning

- Learns from historical data
- Handles large datasets efficiently
- Improves decision-making
- Provides automated predictions

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# 17. LIMITATIONS

Despite good performance, the project has some limitations.

### Limitations

- Prediction depends on dataset quality.
- Limited dataset size may affect generalization.
- Real-time medical data is not included.
- Accuracy may vary for different datasets.
- Medical diagnosis still requires doctor verification.

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# 18. FUTURE ENHANCEMENTS

Future improvements can make the project more advanced.

### Future Scope

- Use deep learning algorithms.
- Deploy the model as a web application.
- Integrate with hospital databases.
- Add real-time patient monitoring.
- Improve accuracy using larger datasets.
- Build mobile healthcare applications.

### Technologies for Future Development

- Flask
- Django
- Streamlit
- TensorFlow
- Cloud Deployment

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# 19. CONCLUSION

The project “Heart Disease Prediction Using Machine Learning” successfully demonstrates the use of machine learning in healthcare analytics.

The dataset was analyzed and preprocessed using various data analysis techniques. Outlier handling, feature scaling, and exploratory data analysis improved the quality of the dataset.

Machine learning algorithms were trained and evaluated for predicting heart disease. The project proves that machine learning can support medical diagnosis systems and help in early disease prediction.

This project also helped in understanding important concepts such as:

- Data preprocessing
- Feature engineering
- Exploratory data analysis
- Model training
- Model evaluation
- Prediction systems

The project can be further enhanced into a real-time healthcare application.

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# 20. REFERENCES


1. Kaggle Heart Disease Dataset
2. Python Documentation
3. Scikit-learn Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Seaborn Documentation
7. Machine Learning Research Articles
8. Healthcare Analytics Resources


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# APPENDIX

## Libraries Used

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
```

```python
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
```

### Sample Workflow

```python
# Load Dataset
import pandas as pd

df = pd.read_csv('heart-2.csv')

# Check Missing Values
print(df.isnull().sum())

# Correlation Matrix
print(df.corr())

# Feature Scaling
from sklearn.preprocessing import StandardScaler

# Train-Test Split
from sklearn.model_selection import train_test_split

# Logistic Regression Model
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
```

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# PROJECT SUMMARY

This mini project focuses on predicting heart disease using machine learning techniques. The project involves preprocessing medical data, handling outliers, visualizing patterns, scaling features, building classification models, and evaluating prediction performance.

The project highlights the practical application of machine learning in healthcare and demonstrates how predictive analytics can improve medical decision-making.

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